a CPU, GPU, Digital Signal Processors (DSP), connectivity controllers, display adapter (different from GPU), GPS processors and a camera input processor on a single chip. This reduces the space these components would otherwise take up and improves overall performance since there are no external controllers needed to control the data flow and conversion. The next time you're in a conversation about how smart phones now are and the many amazing tasks they can perform despite being so slim, think about SoCs.

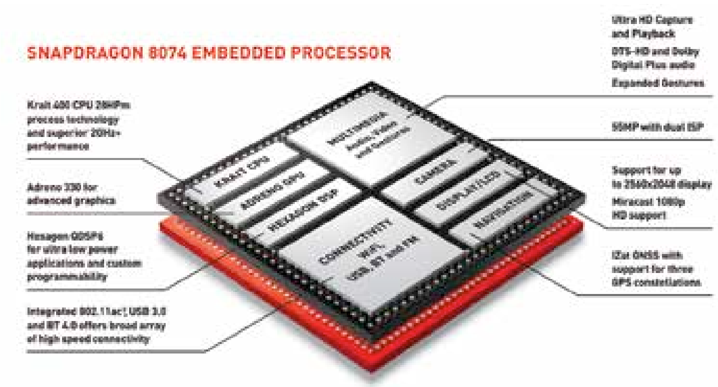
The biggest drawback of a SoC is that since all of the circuitry is contained in a single chip, if say the communications controller were to get damaged, it would require the entire chip to be replaced.

RAM

In most cases, the RAM available on our laptops and desktops wouldn't fit embedded systems for two reasons: firstly, there are no ports to which you could add RAM to, and secondly, since embedded systems need to have everything as compact as possible, the RAMs from desktops/laptops wouldn't fit the criteria. Hence, the RAM too, is built directly on the hardware board. Depending on the need, this RAM needs to be fast and have adequate data storing ability.

Sensors and communication devices

An embedded system can be used to perform either very simple tasks or complex ones. Depending on the variety and number of tasks the system is designed to execute, it would need a corresponding number of sensors. A washing machine panel controller might not need many sensors and can



Snapdragon SoC